

Supplementary appendix

Post-transplant cyclophosphamide as graft-versus-host disease prophylaxis  
after allogeneic haematopoietic stem-cell transplantation: a systematic  
review and Bayesian meta-analysis of infection risk

Marta Stanzani      Dimitrios P Kontoyiannis      Russell E Lewis

This appendix forms part of the article and has not been edited by the journal. It is  
referenced in the main text as “the appendix”.

All results reported in this appendix are from the cohort-deduplicated (Set B) analysis,  
in which one publication per cohort contributes to each outcome.

**S1. Search strategy**

Table 1: Search parameters.

| Field               | Content                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------|
| Databases searched  | PubMed/MEDLINE, EMBASE (Elsevier),<br>SCOPUS                                                    |
| Search dates        | Database inception to [FINAL SEARCH<br>DATE — to be taken from the<br>Covidence/RIS export log] |
| Records retrieved   | PubMed 1061; EMBASE 2603; SCOPUS 481<br>(total 4145)                                            |
| Date limits applied | [to be confirmed — sources conflict]                                                            |

| Field                                 | Content                                                                                                                                                                                           |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Language limit                        | English                                                                                                                                                                                           |
| Publication types excluded via filter | None. Conference abstracts, editorials, letters, and case reports were excluded at screening rather than by search filter, to minimise inappropriate exclusion                                    |
| Other sources searched                | Reference list screening of included studies and prior systematic reviews; forward citation tracking (Web of Science) for key index publications; PROSPERO search for related and ongoing reviews |
| Deduplication and management          | RIS export from each database, imported into Covidence for deduplication and screening                                                                                                            |
| Search piloted against                | Eight known included publications (Holtan 2022, Moiseev 2016, Tang 2023, Papanicolaou 2023, Salas 2023, Massoud 2021, Meyer 2024, Shaffer 2024) to verify recall                                  |

11 The protocol specifies three concept blocks combined with AND, each block an OR of  
12 controlled vocabulary terms and free-text synonyms.

### 13 PubMed/MEDLINE

14 #1 Allogeneic HCT population

15 ("Hematopoietic Stem Cell Transplantation"[Mesh] OR "Bone Marrow Transplantation"

16 OR "allogeneic stem cell transplant\*" [tiab] OR "allo-HCT" [tiab] OR "allo-  
 17 HSCT" [tiab]  
 18 OR "haploidentical" [tiab] OR "HLA-matched" [tiab] OR "matched unrelated donor" [tiab]  
 19 OR "mismatched unrelated donor" [tiab])  
 20  
 21 #2 PTCy intervention  
 22 ("post-transplant cyclophosphamide" [tiab] OR "post-transplantation cyclophosphamide"  
 23 OR "posttransplant cyclophosphamide" [tiab] OR "posttransplantation cyclophosphamide"  
 24 OR "PTCy" [tiab] OR "PT-Cy" [tiab] OR "PTCY" [tiab]  
 25 OR ("Cyclophosphamide" [Mesh] AND "Graft vs Host Disease" [Mesh]))  
 26  
 27 #3 Infection outcomes  
 28 ("Infections" [Mesh] OR "Bacterial Infections" [Mesh] OR "Virus Diseases" [Mesh]  
 29 OR "Mycoses" [Mesh] OR "Bacteremia" [Mesh] OR "Sepsis" [Mesh]  
 30 OR "Cytomegalovirus Infections" [Mesh] OR "Invasive Fungal Infections" [Mesh]  
 31 OR "bacterial infection\*" [tiab] OR "viral infection\*" [tiab] OR "fungal infection"  
 32 OR "BSI" [tiab] OR "sepsis" [tiab] OR "CMV" [tiab] OR "EBV" [tiab] OR "HHV-  
 33 6" [tiab]  
 34 OR "BK virus" [tiab] OR "aspergillosis" [tiab] OR "candida" [tiab] OR "mucormycosis"  
 35 OR "mold" [tiab] OR "yeast" [tiab] OR "mortality" [tiab])  
 36  
 37 #4 #1 AND #2 AND #3  
 38 Filters: Humans; English

39 **EMBASE (Emtree)**

40 **#1 Allogeneic HCT population**

41 ('bone marrow transplantation'/exp OR 'stem cell transplantation'/exp  
42 OR 'allogeneic stem cell transplant\*':ti,ab OR 'allo-hct':ti,ab OR 'allo-  
43 hsct':ti,ab  
44 OR haploidentical:ti,ab OR 'hla-matched':ti,ab OR 'matched unrelated donor':ti,a  
45 OR 'mismatched unrelated donor':ti,ab)

46

47 **#2 PTCy intervention**

48 ('post-transplant cyclophosphamide':ti,ab OR 'post-transplantation cyclophosphami  
49 OR 'posttransplant cyclophosphamide':ti,ab OR 'posttransplantation cyclophospham  
50 OR ptcy:ti,ab OR 'pt-cy':ti,ab  
51 OR ('cyclophosphamide'/exp AND 'graft versus host reaction'/exp))

52

53 **#3 Infection outcomes**

54 ('infection'/exp OR 'bacterial infection'/exp OR 'virus infection'/exp OR 'mycosi  
55 OR 'bacteremia'/exp OR 'sepsis'/exp OR 'cytomegalovirus infection'/exp  
56 OR 'invasive fungal infection'/exp  
57 OR 'bacterial infection\*':ti,ab OR 'viral infection\*':ti,ab OR 'fungal infection  
58 OR bsi:ti,ab OR sepsis:ti,ab OR cmv:ti,ab OR ebv:ti,ab OR 'hhv-  
59 6':ti,ab  
60 OR 'bk virus':ti,ab OR aspergillosis:ti,ab OR candida:ti,ab OR mucormycosis:ti,a  
61 OR mold:ti,ab OR yeast:ti,ab OR mortality:ti,ab)

62

63 #4 #1 AND #2 AND #3 AND [humans]/lim AND [english]/lim

64 SCOPUS

65 TITLE-ABS-KEY(

66 ("allogeneic stem cell transplant\*" OR "allo-HCT" OR "allo-HSCT"

67 OR "hematopoietic stem cell transplantation" OR "bone marrow transplantation"

68 OR haploidentical OR "HLA-matched" OR "matched unrelated donor"

69 OR "mismatched unrelated donor")

70 AND

71 ("post-transplant cyclophosphamide" OR "post-transplantation cyclophosphamide"

72 OR "posttransplant cyclophosphamide" OR "posttransplantation cyclophosphamide"

73 OR PTCy OR "PT-Cy" OR (cyclophosphamide AND "graft versus host disease"))

74 AND

75 (infection\* OR "bacterial infection\*" OR "viral infection\*" OR "fungal infection\*" OR

76 OR BSI OR sepsis OR bacteremia OR CMV OR "cytomegalovirus infection\*" OR EBV

77 OR "HHV-6" OR "BK virus" OR aspergillosis OR candida OR mucormycosis OR mold

78 OR yeast OR "invasive fungal infection\*" OR mortality)

79 ) AND LANGUAGE(english)

## S2. Studies excluded at full-text screening (n=214)

Table 2: Reasons for exclusion at full-text screening.

| Exclusion reason                   | n          |
|------------------------------------|------------|
| Conference abstract                | 116        |
| Single-arm or no PTCy contrast     | 41         |
| Review, not primary study          | 21         |
| Duplicate publication              | 13         |
| Systematic review or meta-analysis | 11         |
| Non-eligible population            | 5          |
| Protocol or survey only            | 4          |
| Other                              | 3          |
| <b>Total</b>                       | <b>214</b> |

## S3. Characteristics of included studies

Full study-level (table S4a, 251 rows) and arm-level (table S4b, 525 rows) characteristics are provided as supplementary data files (`Table_S4a_study_characteristics.csv`, `Table_S4b_arm_characteristics.csv`), and are reproduced from the extraction database in the accompanying notebook *Study and arm characteristics*.

## 86 S5. Risk of bias

Table 3: ROBINS-I domain-level risk of bias, counts by judgement (n = 211 comparative observational studies; single-arm descriptive studies excluded).

| Domain                                    | Low | Moderate | Serious |
|-------------------------------------------|-----|----------|---------|
| D1 Confounding                            | 0   | 83       | 128     |
| D2 Selection of participants              | 26  | 140      | 45      |
| D3 Classification of interventions        | 51  | 155      | 5       |
| D4 Deviations from intended interventions | 193 | 18       | 0       |
| D5 Missing data                           | 70  | 140      | 1       |
| D6 Measurement of outcomes                | 101 | 101      | 2       |
| D7 Selection of the reported result       | 161 | 42       | 1       |

87 Full justification narratives are available in `rob.csv`, submitted as a supplementary data  
88 file.

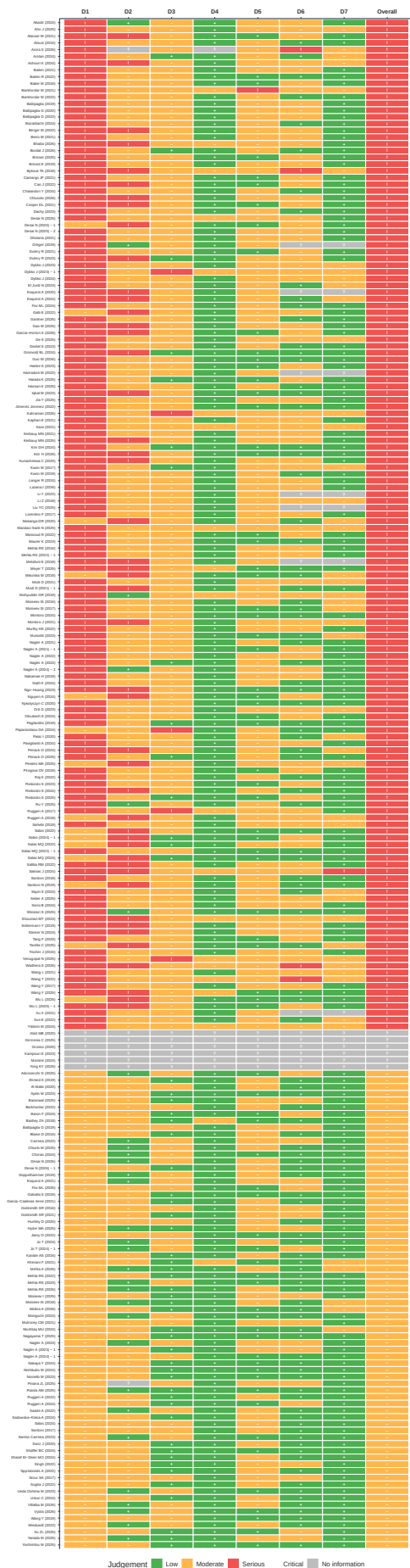


Figure 1: ROBINS-I traffic-light plot for all comparative observational studies.



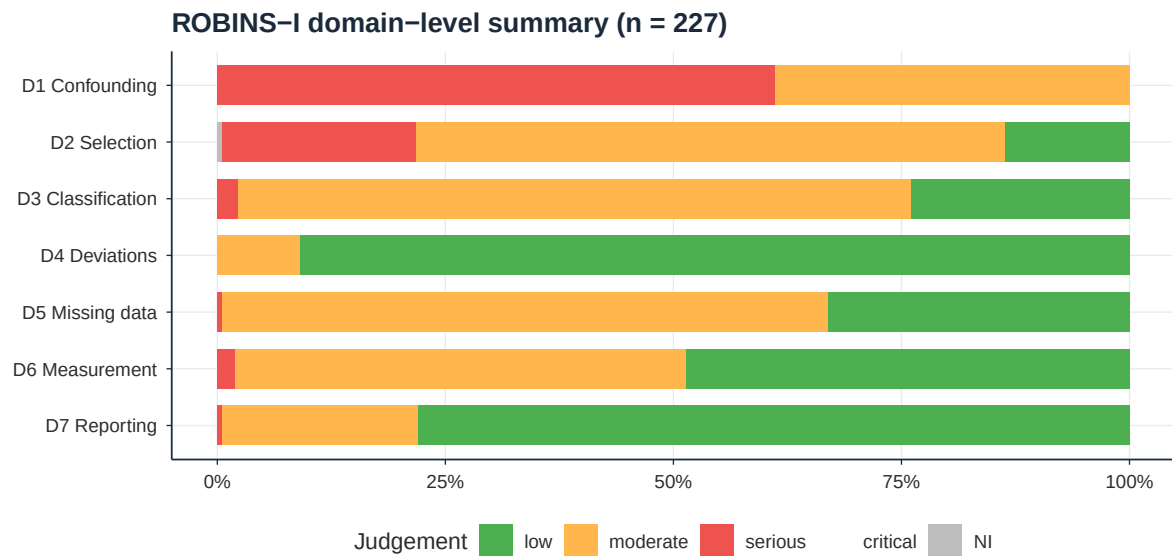


Figure 2: ROBINS-I domain-level summary.

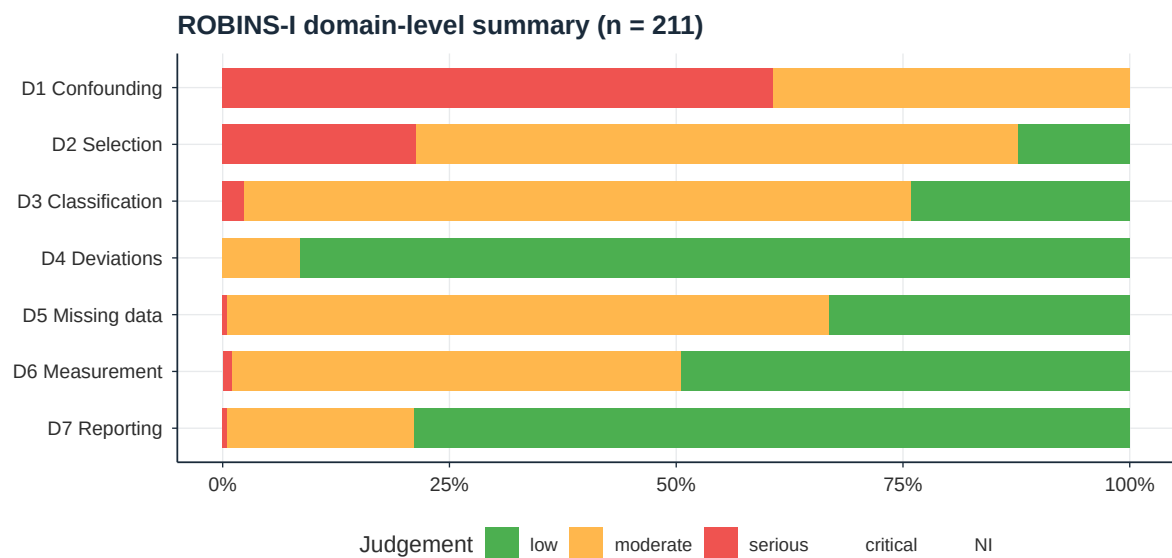


Figure 3: RoB 2 summary for the 14 randomised controlled trials.

## S6. Model specification and priors

Primary model (M1), for study  $s$ , arm  $i$ :

$$\text{events}_i \mid n_i \sim \text{Binomial}\left(n_i, \text{logit}^{-1}\left(\alpha + \beta_{\text{PTCy}}x_i + \beta_{\text{tp}}t_i + u_{s[i]}\right)\right), \quad u_s \sim \mathcal{N}(0, \tau^2)$$

where  $x_i$  indicates a PTCy arm and  $t_i$  indicates a late ( 1 year) rather than early (day +100 or +180) timepoint.

Table 4: Prior specification for the primary model.

| Parameter            | Prior                       |
|----------------------|-----------------------------|
| Fixed effects ( )    | $\text{N}(0, 2 \cdot 5)$    |
| Intercept ( )        | $\text{N}(0, 1 \cdot 5)$    |
| Between-study SD ( ) | Student $t(3, 0, 1)$ , half |

The steroid-mediation model (M2) adds arm-level systemic steroid exposure percentage, mean-centred, as an additional fixed effect.

MCMC: four chains  $\times$  4000 iterations (1000 warmup), `adapt_delta` = 0.95, giving 12 000 post-warmup draws per model.

## S7. MCMC diagnostics

All Bayesian models were fitted with `brms` v2.23.0 and the `rstan` backend.  $\hat{R}$  (threshold  $< 1 \cdot 01$ ), bulk and tail effective sample size (ESS; threshold  $> 400$ ), and the number of divergent transitions are reported for every fitted model. Values are the worst case (maximum  $\hat{R}$ , minimum ESS) across all model parameters.

Table 5: MCMC convergence diagnostics, comparison 1 models.

| Model    | Max $\hat{R}$  | Min bulk ESS | Min tail ESS | Divergences |
|----------|----------------|--------------|--------------|-------------|
| OS M1    | $1 \cdot 0037$ | 1729         | 2962         | 0           |
| OS M2    | $1 \cdot 0022$ | 2645         | 4217         | 0           |
| NRM M1   | $1 \cdot 0017$ | 4176         | 5261         | 0           |
| NRM M2   | $1 \cdot 0017$ | 3061         | 4066         | 0           |
| aGVHD M1 | $1 \cdot 0012$ | 2178         | 3317         | 0           |
| CMV M1   | $1 \cdot 0011$ | 2814         | 4158         | 0           |
| CMV M2   | $1 \cdot 0008$ | 3323         | 4990         | 0           |
| BSI M1   | $1 \cdot 0011$ | 3643         | 5034         | 0           |
| IFI M1   | $1 \cdot 0021$ | 2795         | 4074         | 0           |
| RRM M1   | $1 \cdot 0011$ | 3161         | 3767         | 0           |
| BK M1    | $1 \cdot 0013$ | 3099         | 4143         | 0           |
| IRM M1   | $1 \cdot 0010$ | 2965         | 4257         | 0           |

Table 6: MCMC convergence diagnostics, comparison 2 models.

| Model    | Max $\mathbf{R}$ | Min bulk ESS | Min tail ESS | Divergences |
|----------|------------------|--------------|--------------|-------------|
| OS M1    | $1 \cdot 0030$   | 3185         | 4841         | 0           |
| OS M2    | $1 \cdot 0011$   | 3714         | 5136         | 0           |
| NRM M1   | $1 \cdot 0035$   | 2876         | 2293         | 0           |
| aGVHD M1 | $1 \cdot 0010$   | 3191         | 4855         | 0           |
| CMV M1   | $1 \cdot 0020$   | 3436         | 5356         | 0           |
| CMV M2   | $1 \cdot 0007$   | 3321         | 5570         | 0           |
| RRM M1   | $1 \cdot 0002$   | 3225         | 4933         | 0           |
| BK M1    | $1 \cdot 0592$   | 67           | 17           | 0           |

102 All models except C2 BK achieved  $\mathbf{R} < 1 \cdot 01$  with zero divergent transitions and bulk  
103 ESS above 1000. The C2 BK model ( $k = 3$ ) did not converge to that standard ( $\mathbf{R} =$   
104  $1 \cdot 0592$ , bulk ESS 67, tail ESS 17) and its posterior should be read as indicative only.

105 All models show adequate calibration, with points clustered near the identity line and  
106 symmetric scatter. The C1 IFI model ( $k = 6$ ) shows the widest scatter, consistent with  
107 its large posterior (figure 5) and the heterogeneity identified in S9. No systematic over-  
108 or under-prediction is evident in any model.

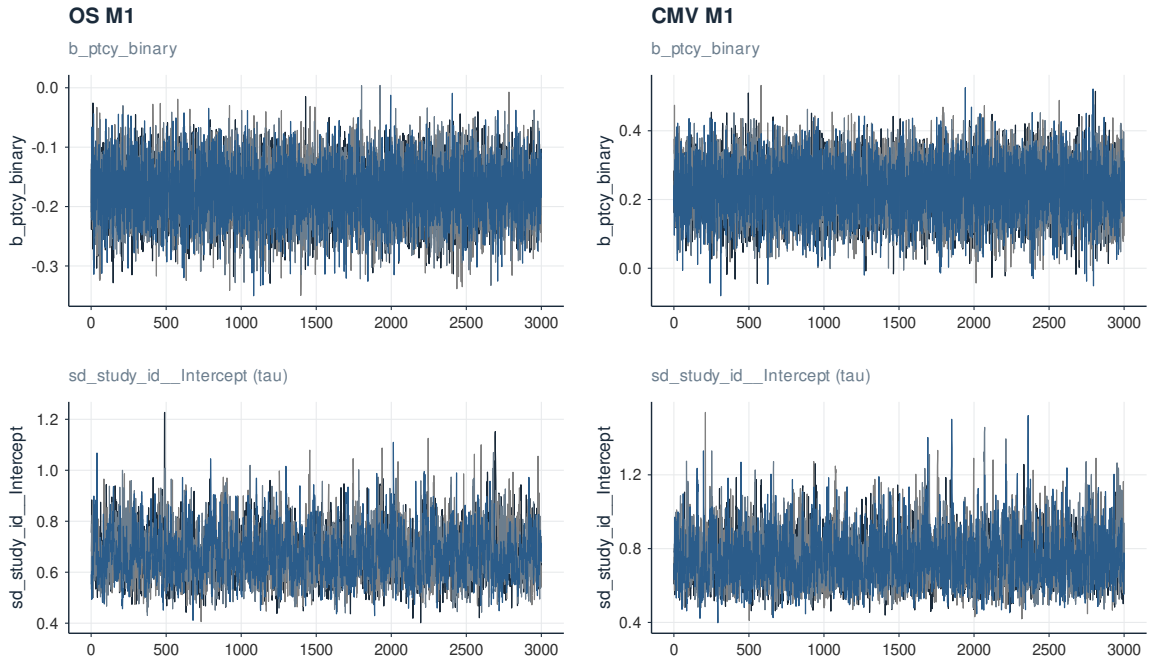


Figure 4: Trace plots for  $b\_ptcy\_binary$  (top) and  $sd\_study\_id\_Intercept(\tau)$  (bottom) in OS M1 (left) and CMV M1 (right). All four chains mix well across the full post-warmup phase.

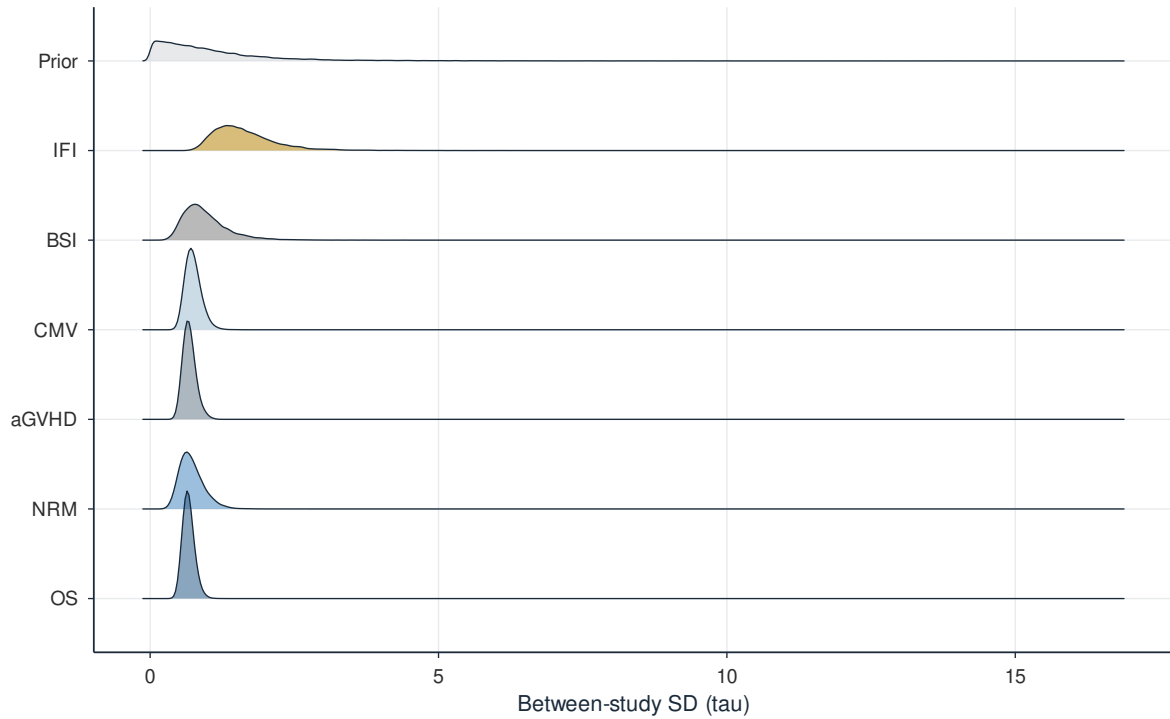


Figure 5: Posterior densities of  $\tau$  (between-study SD) for comparison 1 M1 models. The half-Student  $t(3, 0, 1)$  prior common to all models is shown in grey. IFI shows the widest posterior, consistent with the structural heterogeneity identified in the leave-one-out analysis (S9).

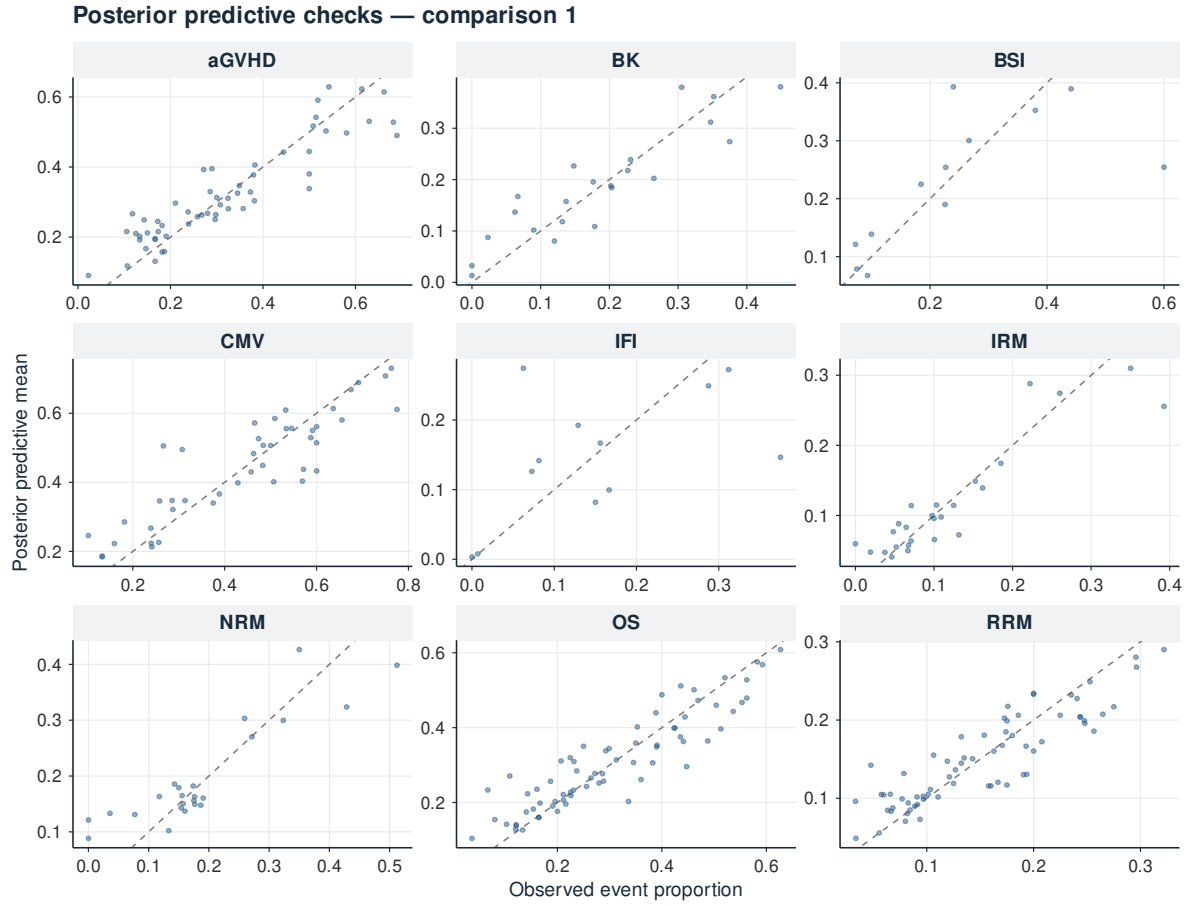


Figure 6: Posterior predictive checks, comparison 1. Each point is one study arm: the observed event proportion against its posterior predictive mean, over 500 posterior draws. The dashed line is the identity.

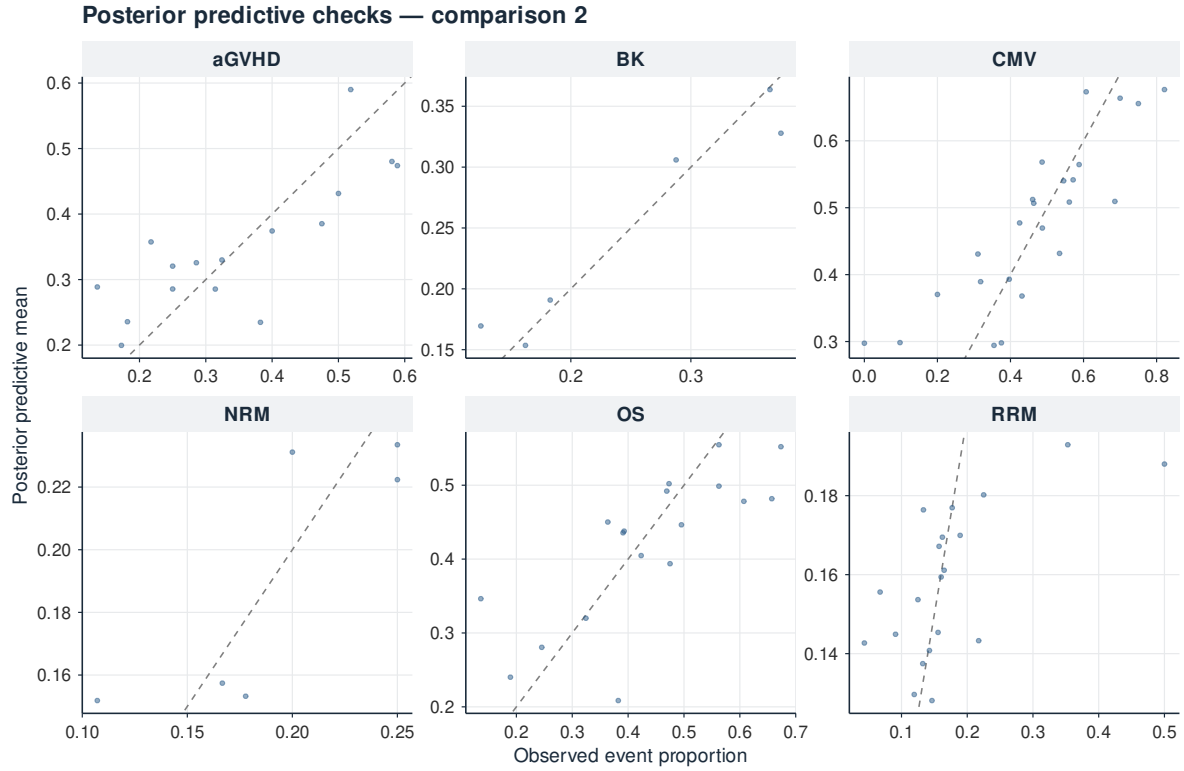


Figure 7: Posterior predictive checks, comparison 2. Layout and interpretation as in figure 6.

## S8. Forest plots

Individual study odds ratios with 95% CIs and the pooled frequentist random-effects estimate (`metafor`, REML), shown alongside the Bayesian estimates reported in the main text as concordance checks. All plots are from the cohort-deduplicated datasets.

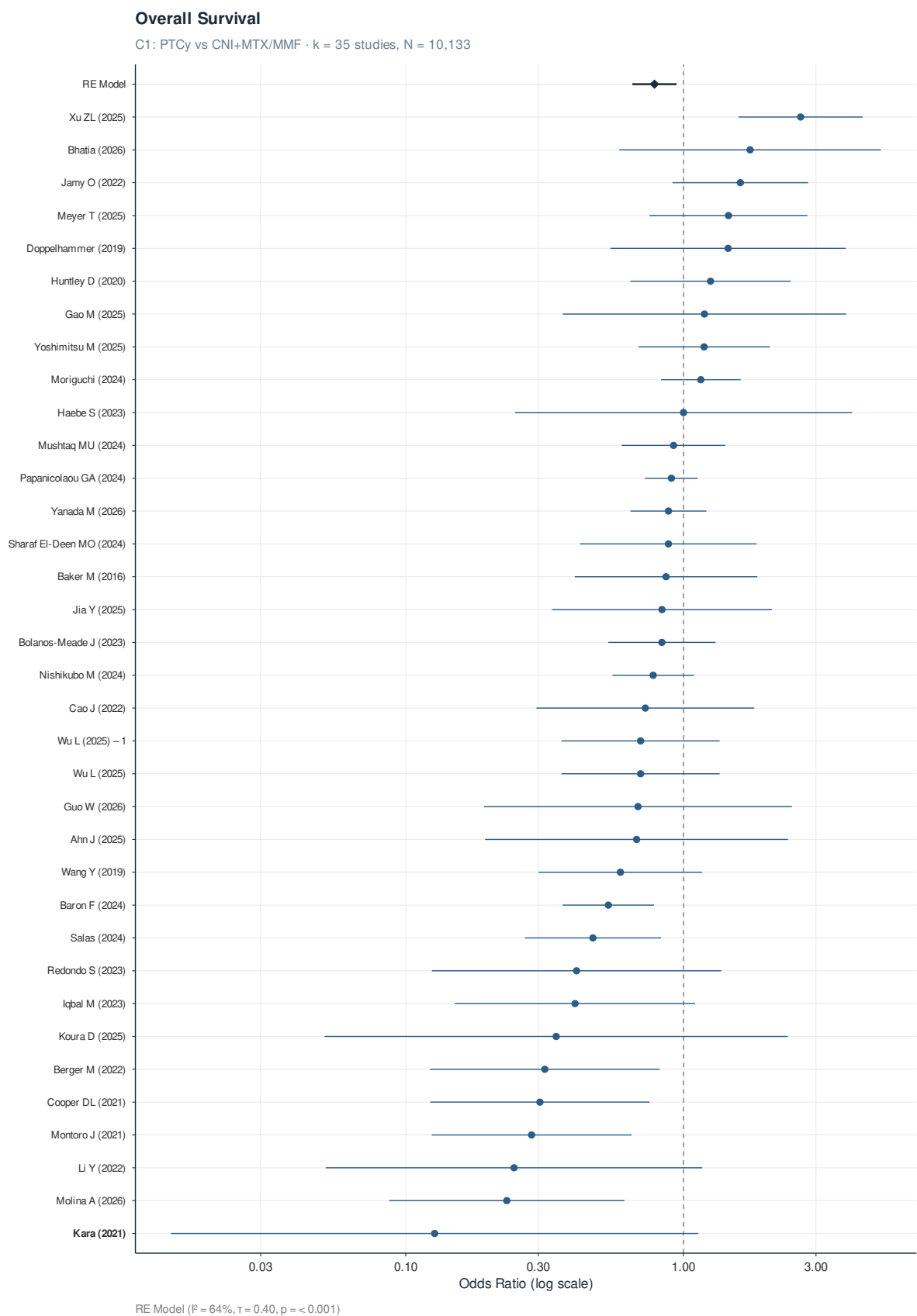
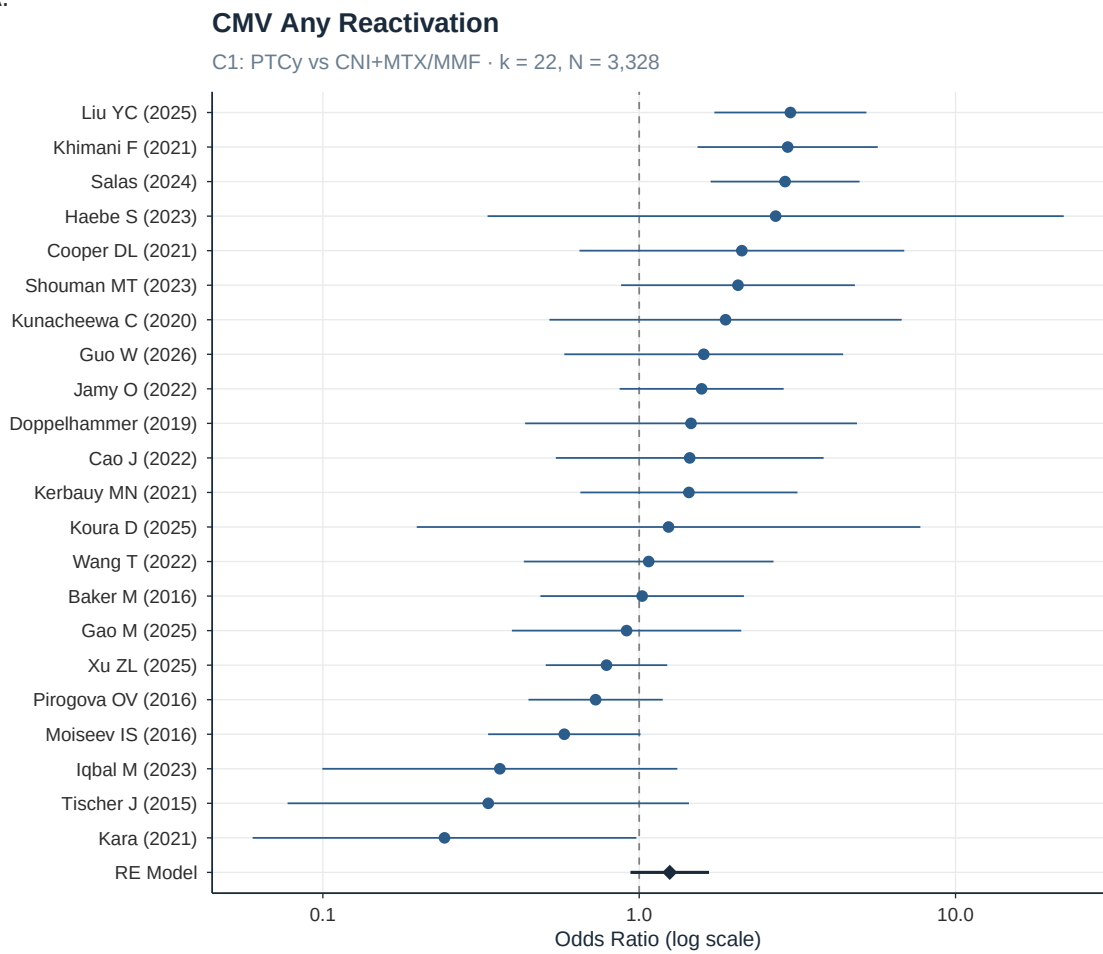


Figure 8: Overall survival, frequentist REML concordance check for figure 2 of the main text.



A.



B.

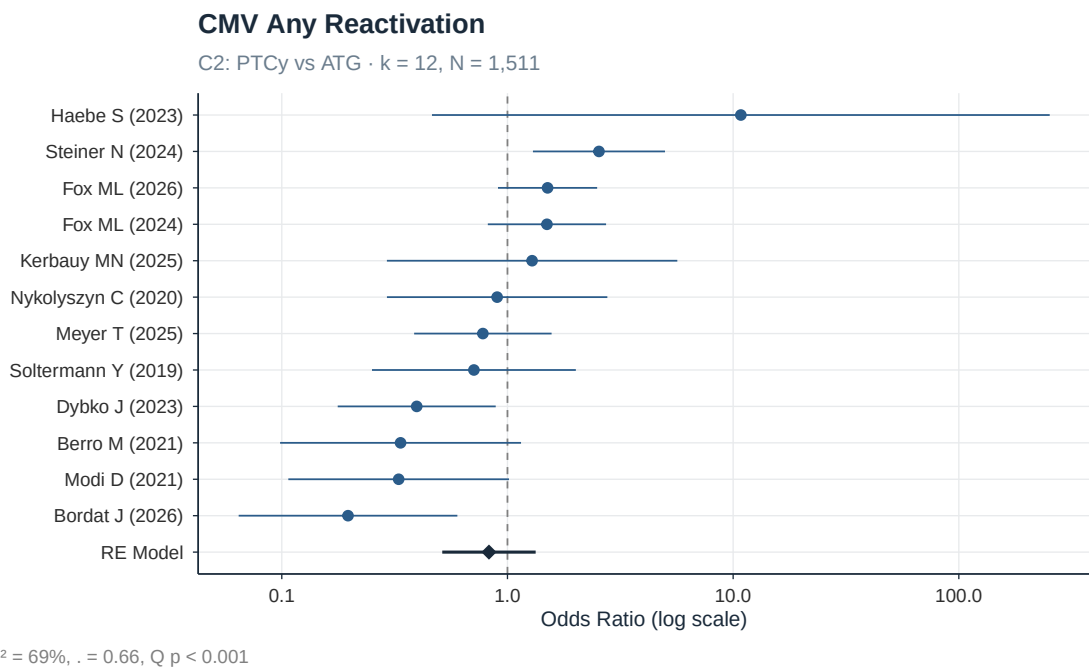
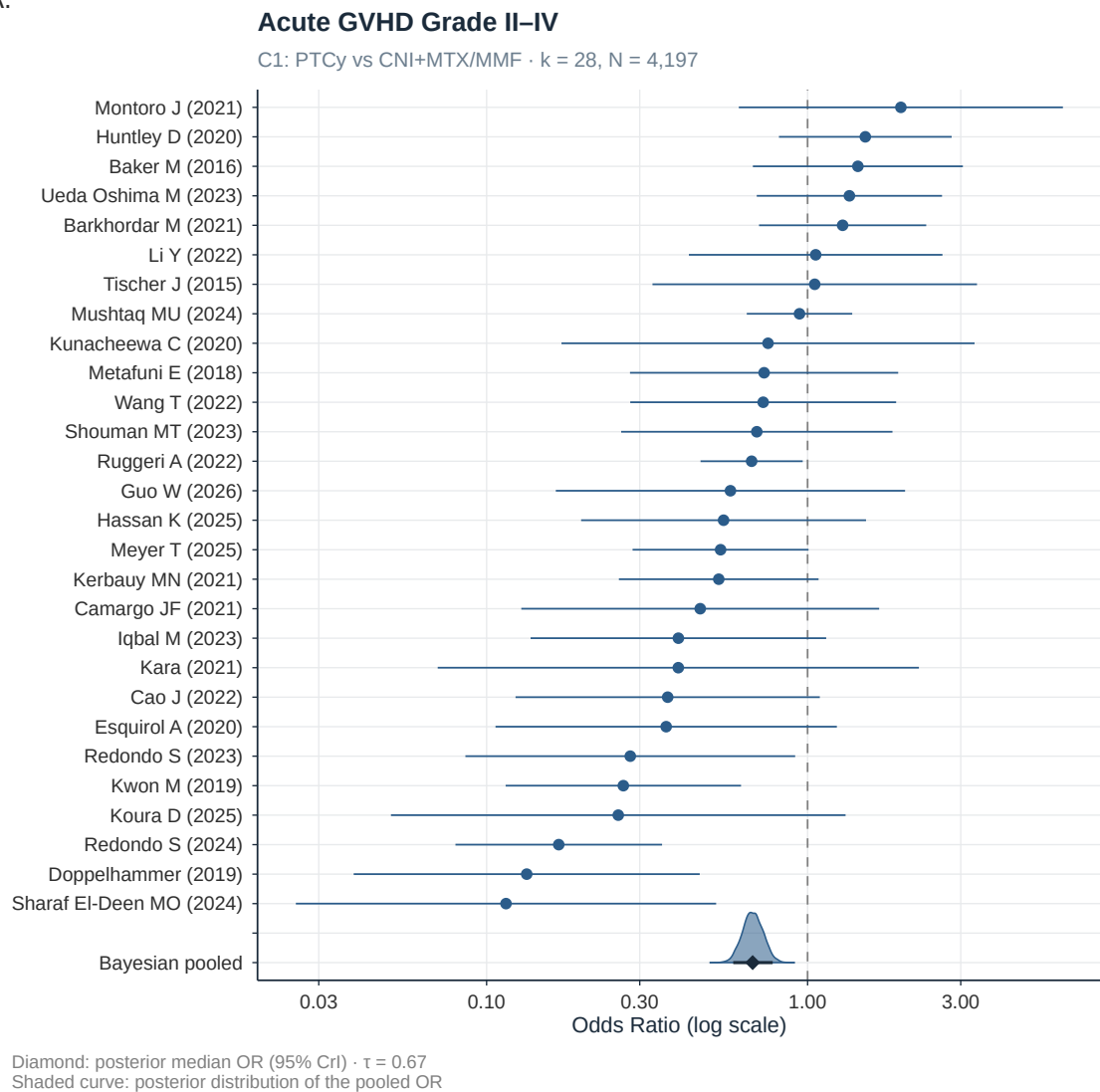


Figure 9: CMV any-reactivation, frequentist REML concordance check for figure 3 of the main text.

A.



B.

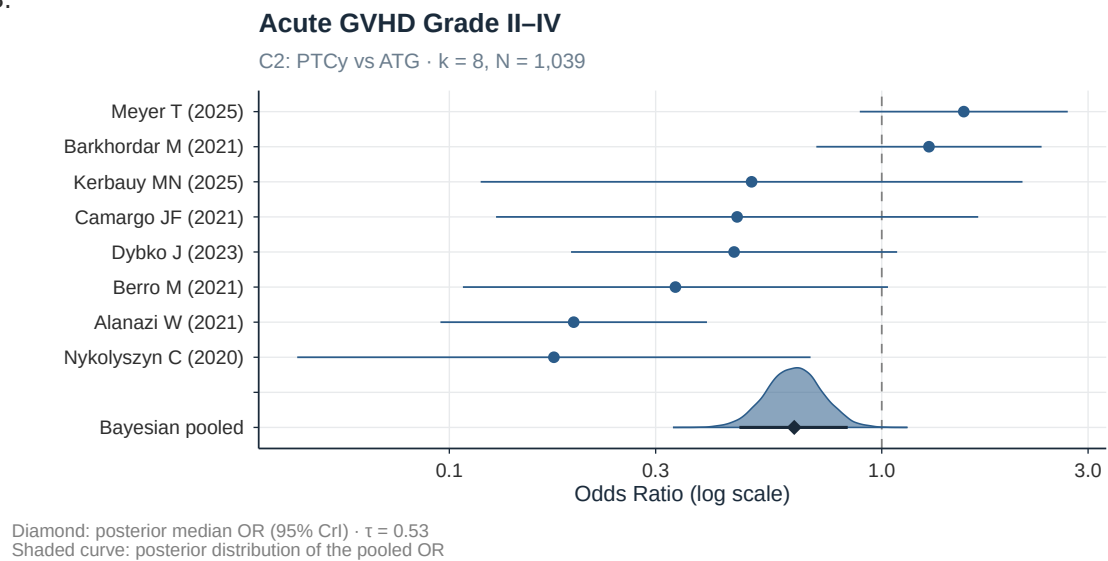
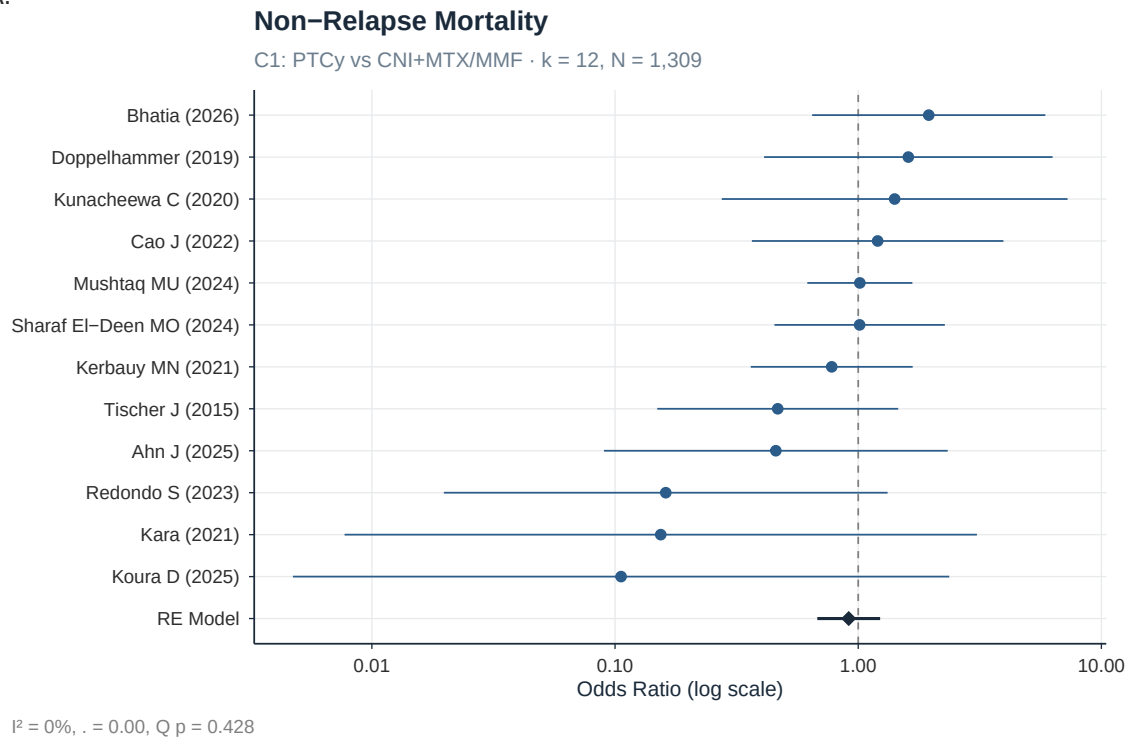


Figure 10: Acute GVHD grade II–IV.

A.



B.

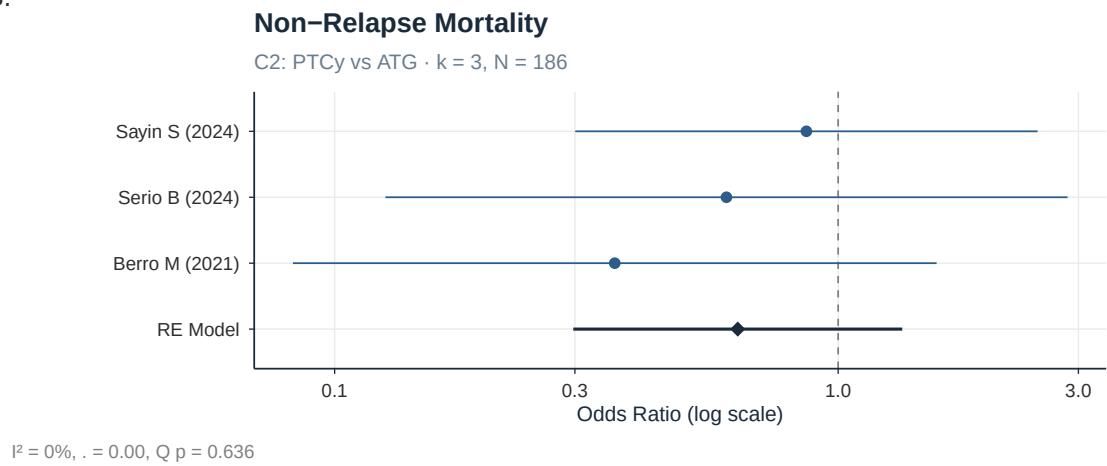


Figure 11: Non-relapse mortality.

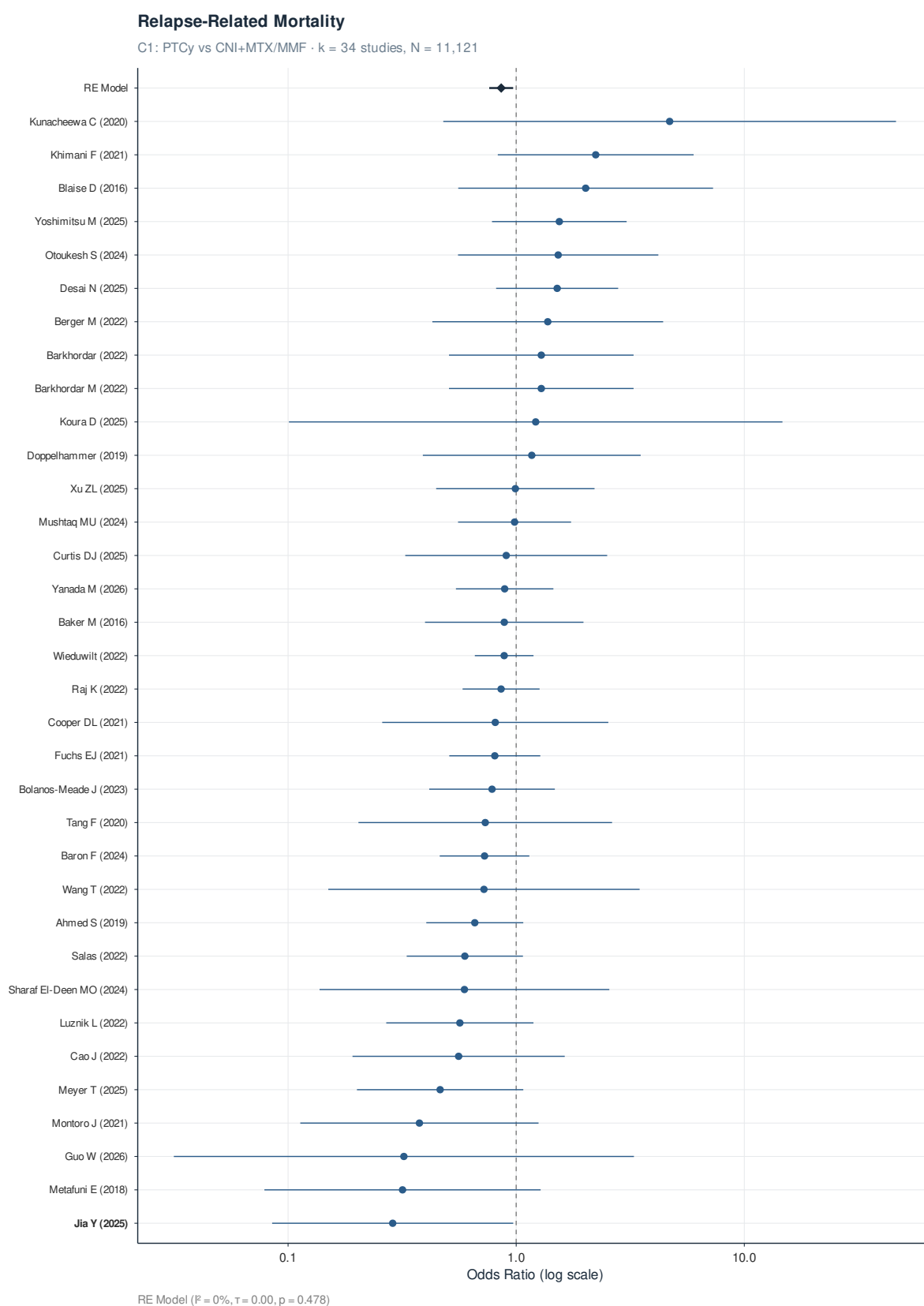


Figure 12: Relapse-related mortality.

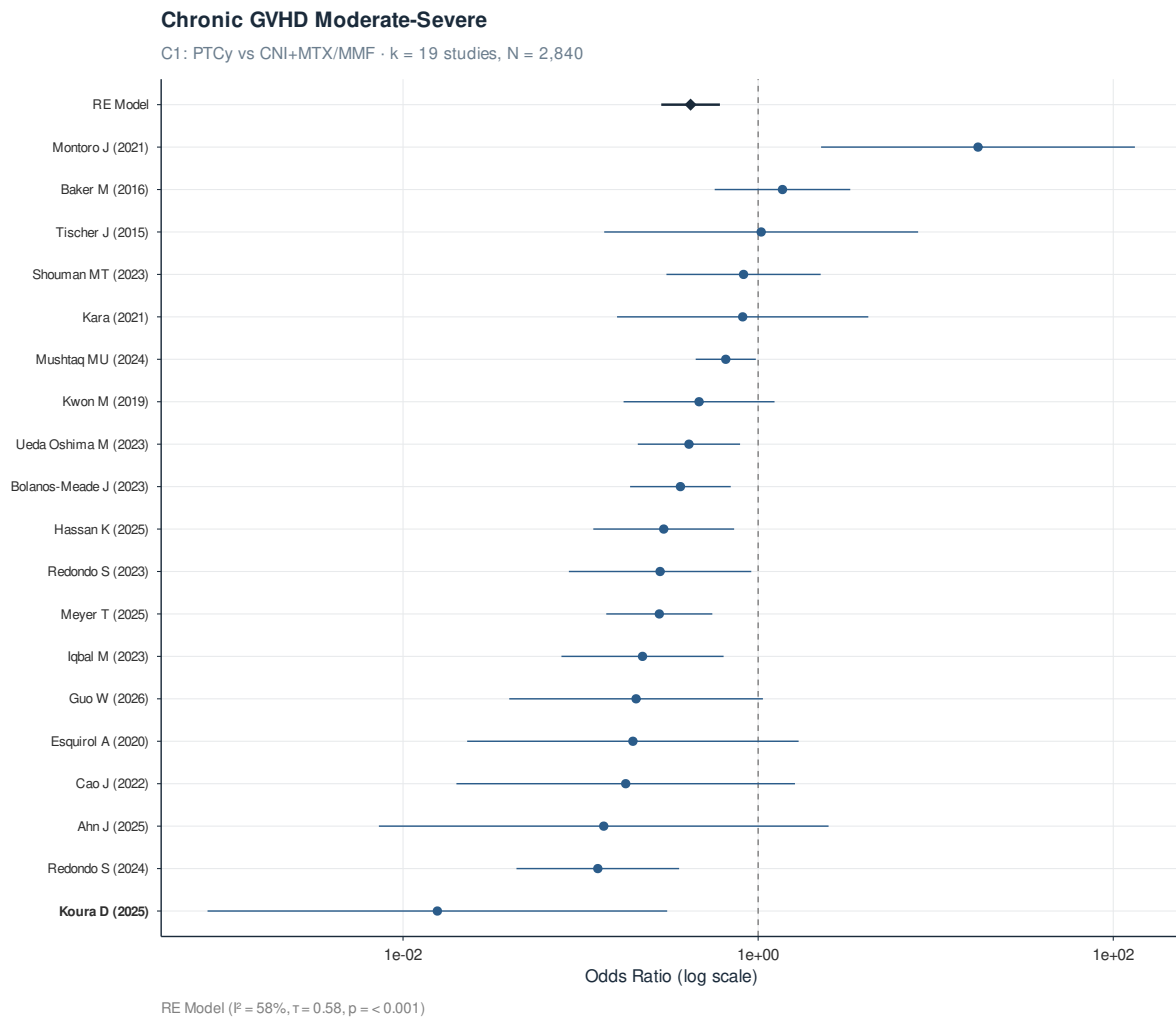


Figure 13: Chronic GVHD, moderate to severe.

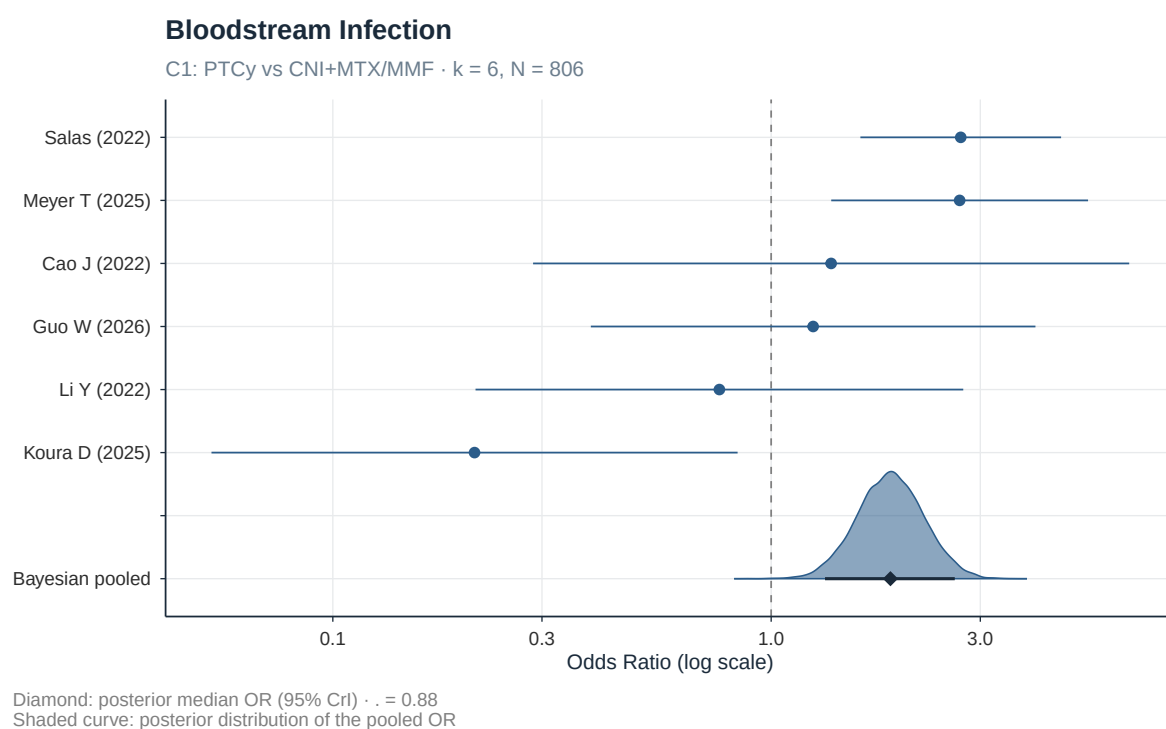


Figure 14: Bloodstream infection, any pathogen.

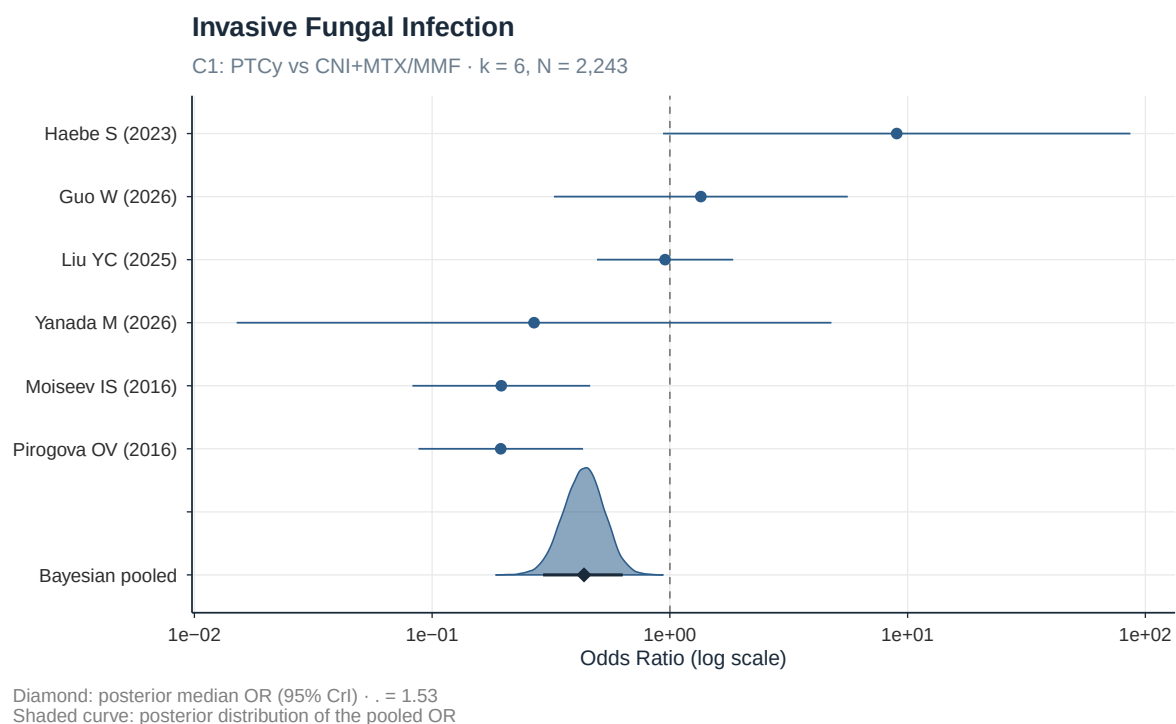
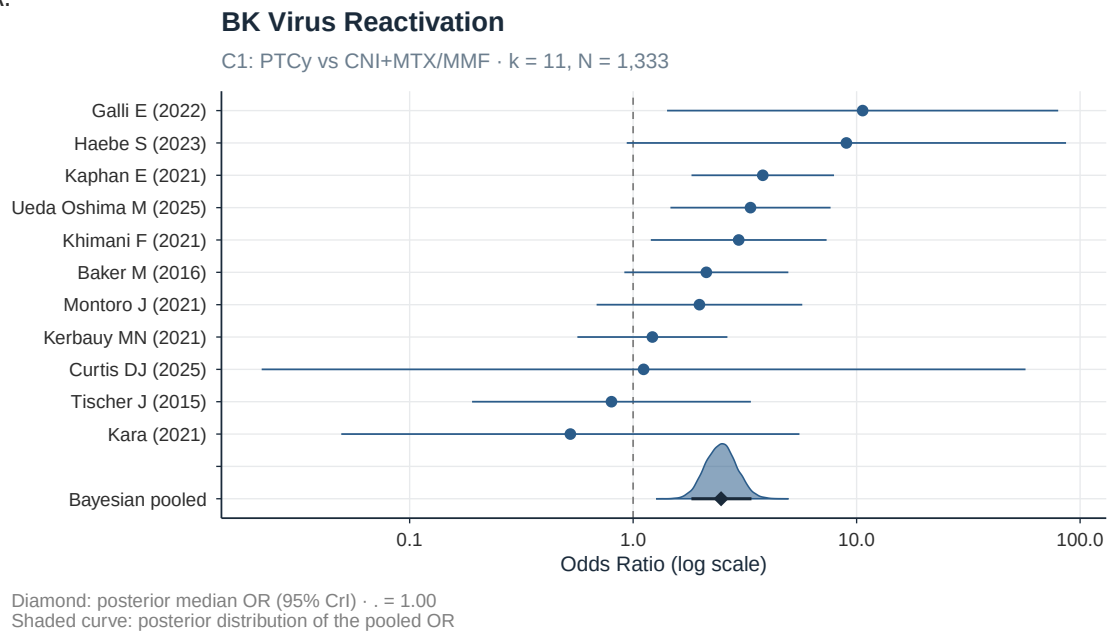


Figure 15: Invasive fungal infection, any.

A.



B.

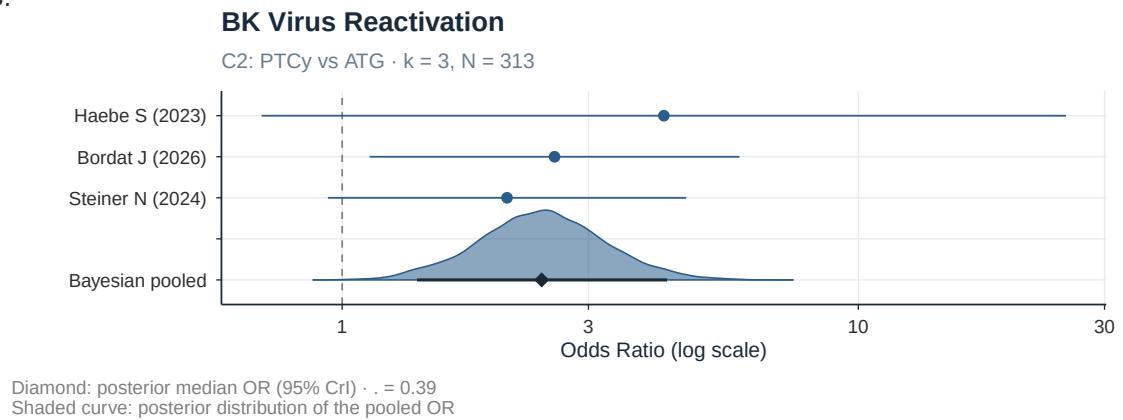


Figure 16: BK virus reactivation.

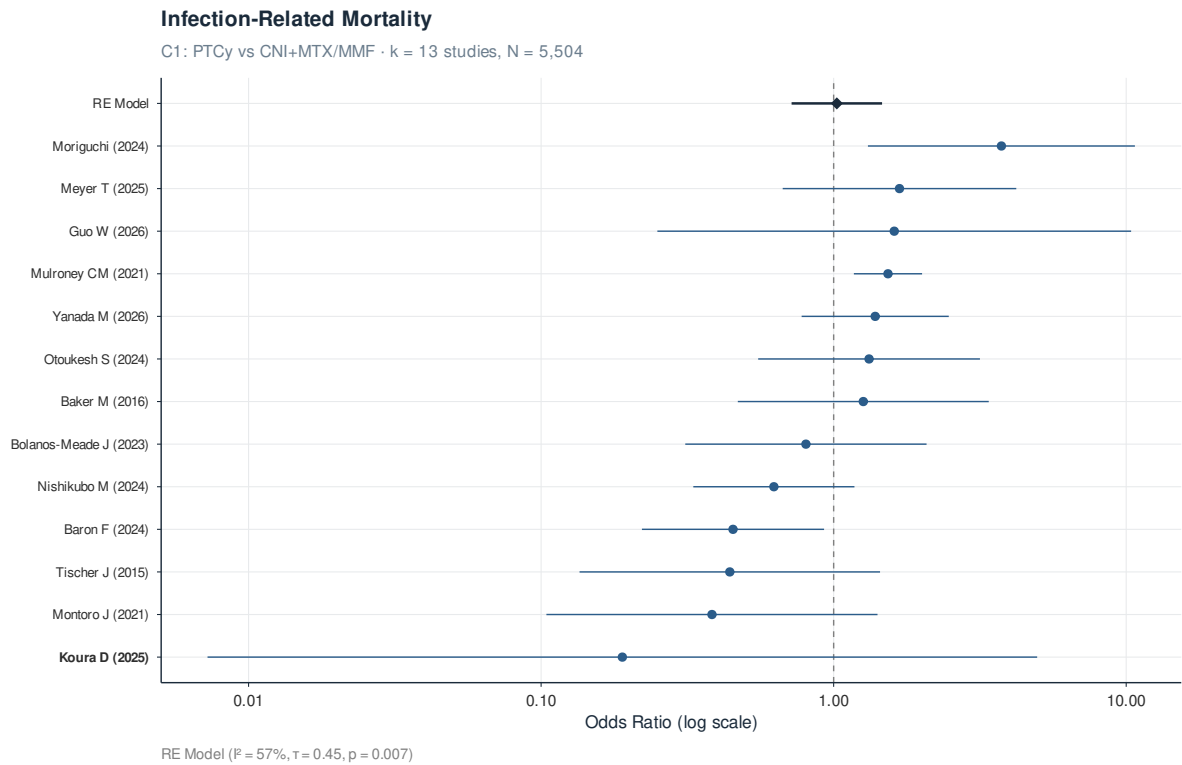


Figure 17: Infection-related mortality.

## S9. Sensitivity analyses

Full frequentist results for all models are provided in `freq_results_setB.csv`, a supplementary data file, which carries the deduplicated fits where deduplication applied and the original fits otherwise (see its `source` column). Direction was concordant between Bayesian and frequentist estimates for every outcome. Two framework-specific differences are worth noting: C1 BSI (Bayesian odds ratio  $1 \cdot 87$  versus frequentist  $1 \cdot 29$  [ $0 \cdot 61$ – $2 \cdot 73$ ]) and C1 IFI ( $0 \cdot 43$  versus  $0 \cdot 60$  [ $0 \cdot 21$ – $1 \cdot 74$ ]), both reflecting the influence of informative priors at small  $k$ .



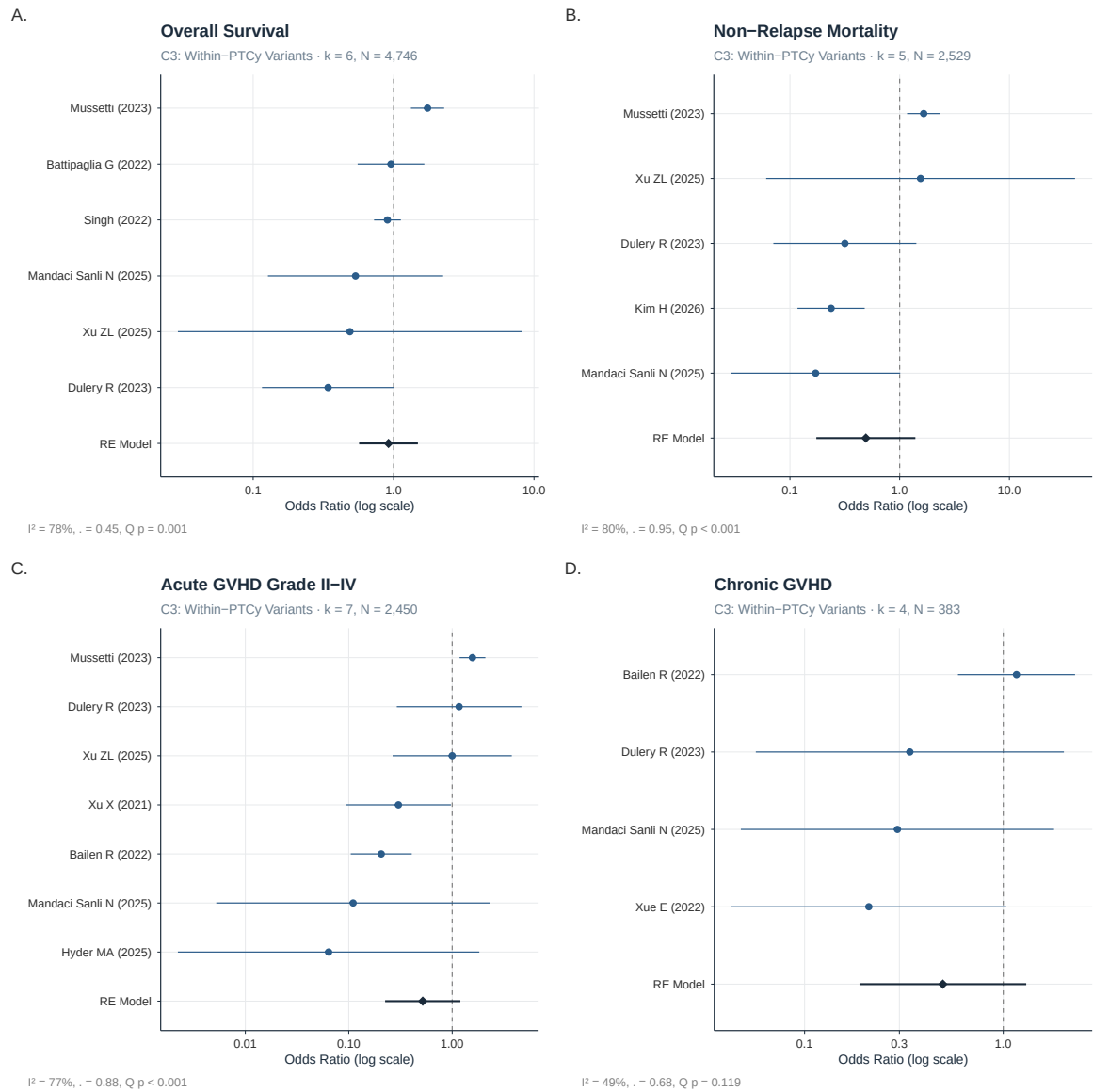


Figure 18: Comparison 3, within-PTCy regimen variants, all outcomes.

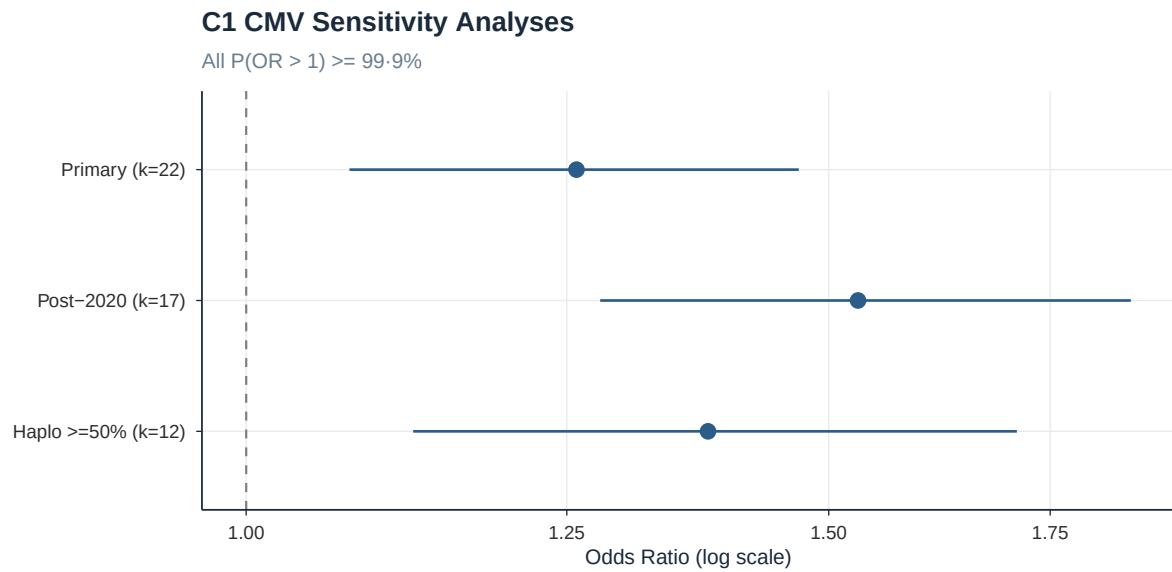


Figure 19: CMV sensitivity models, comparison 1: primary model, post-2020 restriction, and haploidentical-dominant restriction.

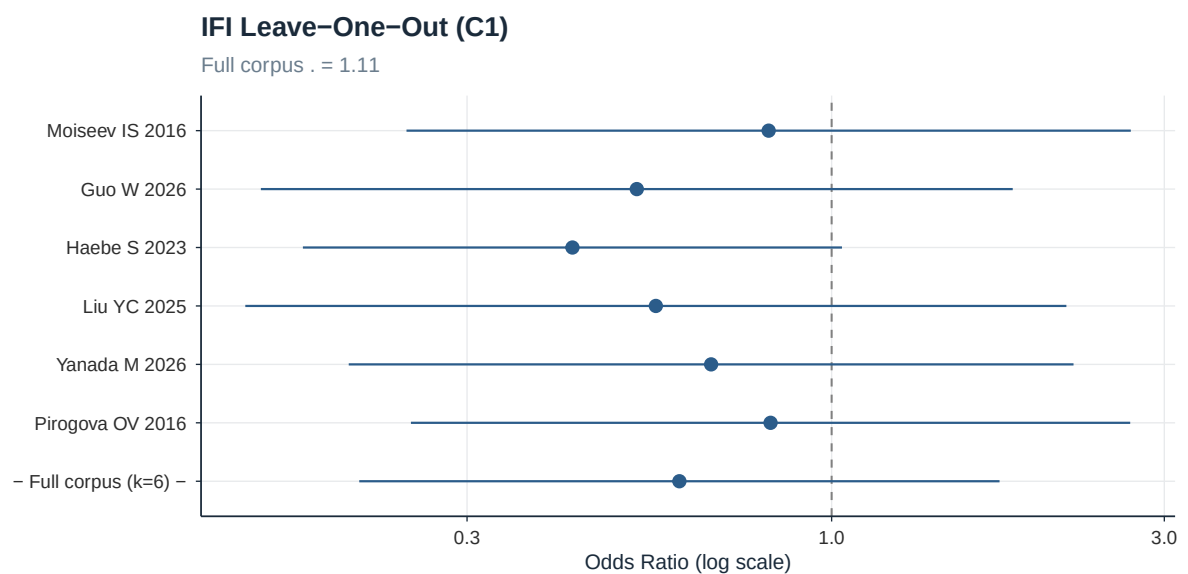


Figure 20: IFI leave-one-out analysis, comparison 1 (frequentist REML).

## S10. Publication bias

Robust Bayesian model-averaging (RoBMA) models were fitted for all outcomes with k = 6.

## S11. GRADE certainty-of-evidence profiles

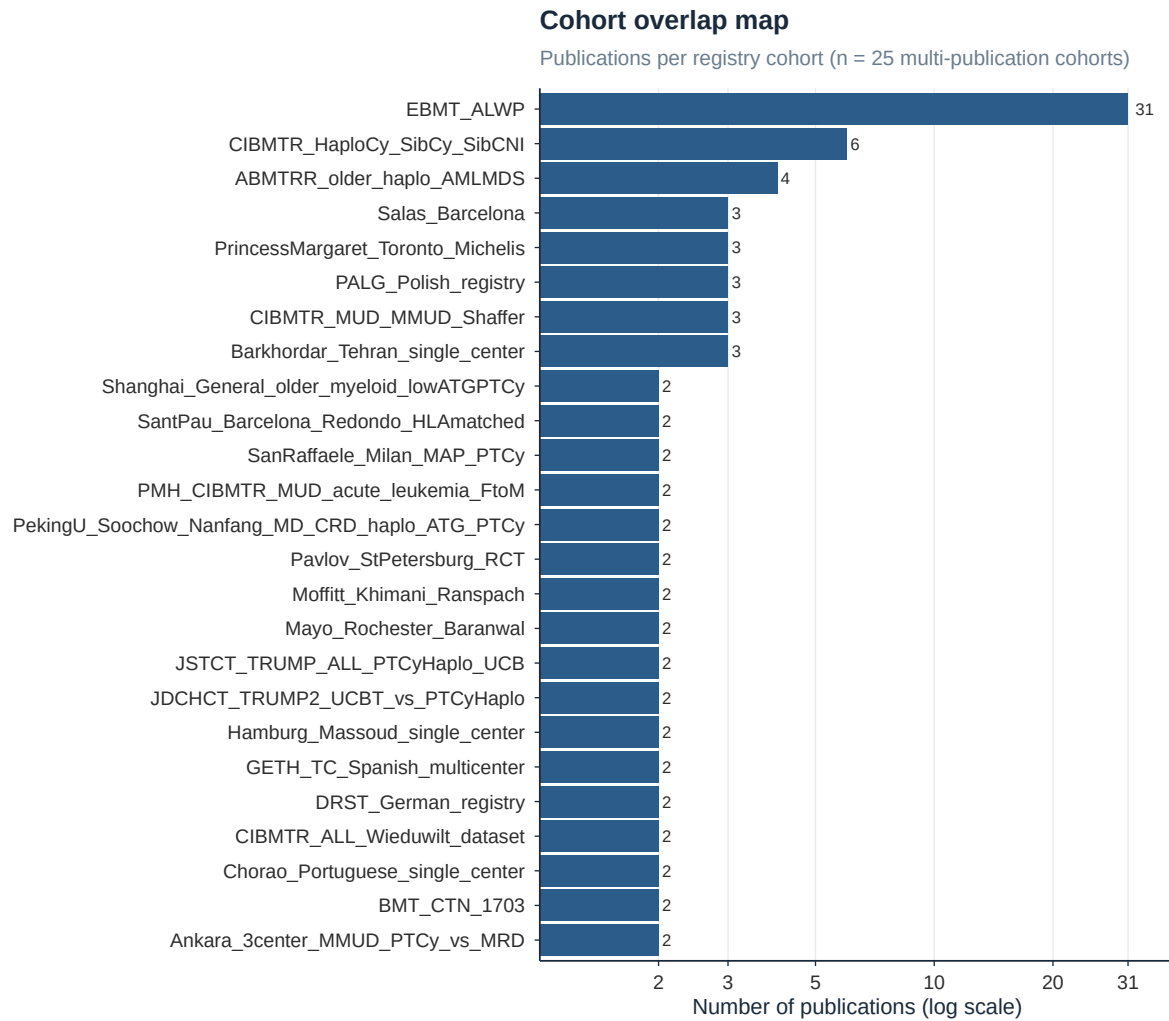


Figure 21: GRADE certainty-of-evidence profiles for the pre-specified primary and secondary outcomes.

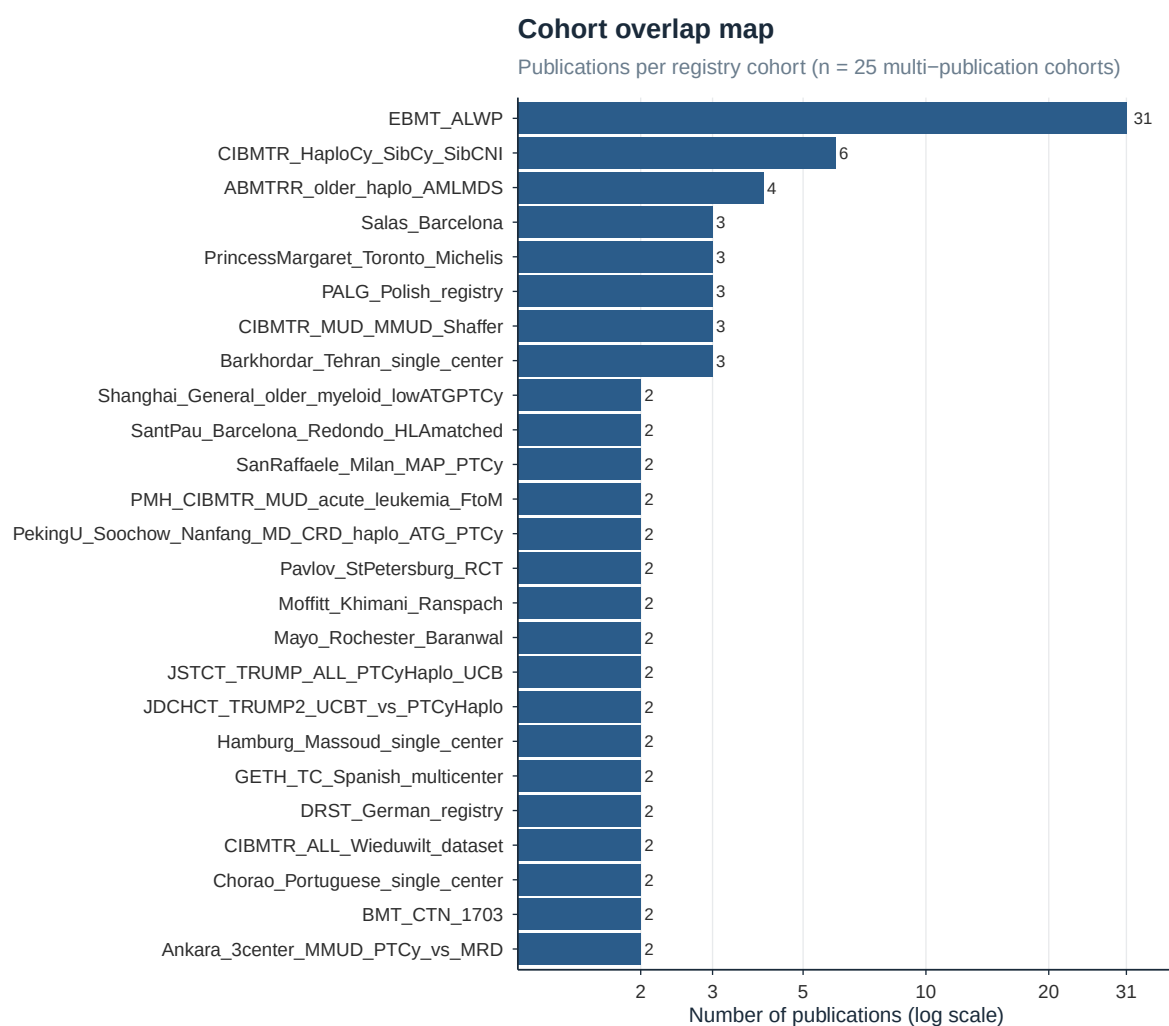


Figure 22: Cohort overlap map. Studies sharing patient cohorts were flagged and the most comprehensive publication designated as primary for each cohort–outcome combination (Set B, the adopted analysis).

## 125 **S12. Cohort overlap**

## 126 **S13. PROSPERO registration**

127 The study protocol was registered with PROSPERO (CRD420261372156) before data  
128 extraction. Protocol amendments were: (1) a Bayesian rather than frequentist primary  
129 analysis framework; (2) addition of BK virus reactivation, infection-related mortality,  
130 and relapse-related mortality as secondary outcomes; and (3) addition of the steroid-  
131 mediation (M2) model.

## 132 **S14. PRISMA 2020 checklist**